

The Constancy Theorem for Currents

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Abstract

We introduce distributions and currents on general Banach spaces and prove some of their basic properties. We then prove the constancy theorem, which characterizes currents whose boundary is restricted to some subset of $\mathbb{R}^n$.

1. Introduction

Differential forms provide methods of integration on oriented manifolds that go beyond what can be done through integration with respect to measures. In turn, currents generalize the notion of differential forms and yield applications to integration over rectifiable sets. Introduced by Schwartz and de Rham, the theory of currents, especially normal, rectifiable, and integral currents, was further developed by Federer in his study of what he called homological integration.

The following definitions are as in [Fed96, Chapter 4.1.1]: We denote by $\wedge_m V$ the space of m -vectors of V and $\wedge^m V$ the space of m -covectors, which is identified with $\text{Hom}(\wedge_m V, \mathbb{R})$. We refer the reader to [Fed96, Chapter 1.3-1.4] for their construction.

Definition 1.1. Let X and Y be Banach spaces such that $\dim X < \infty$, and let U be an open subset of X . We define $\mathcal{E}(U, Y)$ to be the space of all class ∞ , or infinitely differentiable, functions from U to Y .

Definition 1.2. We also define a family of seminorms as follows: for every nonnegative integer i and compact subset K of U , let

$$\nu_K^i(\phi) = \sup\{\|D^j \phi(x)\| : 0 \leq j \leq i, x \in K\}$$

for all $\phi \in \mathcal{E}(U, Y)$, where D^j is the j th differential and $\|D^j \phi(x)\|$ denotes the operator norm.

This family induces a Hausdorff topology on $\mathcal{E}(U, Y)$ where the neighborhoods of $\phi \in \mathcal{E}(U, Y)$ are the sets

$$\{\psi \in \mathcal{E}(U, Y) : \nu_K^i(\psi - \phi) < r\}$$

for $r > 0$.

Definition 1.3. Let $\mathcal{E}'(U, Y)$ be the vector space of continuous linear functions from $\mathcal{E}(U, Y)$ to $\mathbb{R}$. We will use $\langle \cdot, \cdot \rangle$ to denote the dual pairing on $\mathcal{E}'(U, Y) \otimes \mathcal{E}(U, Y)$ given by evaluation (note that the order is reversed from the notation in [Fed96]).

Using the above definitions, we now define the support of elements of $\mathcal{E}(U, Y)$ and $\mathcal{E}'(U, Y)$, noting that the support of an element of $\mathcal{E}'(U, Y)$ is defined on U rather than $\mathcal{E}(U, Y)$.

Definition 1.4. For $\phi \in \mathcal{E}(U, Y)$, we define

$$\text{spt } \phi = U \setminus \bigcup \{W : W \text{ is open, } \phi(x) = 0 \text{ for all } x \in W\},$$

and similarly for $T \in \mathcal{E}'(U, Y)$, we define

$$\text{spt } T = U \setminus \bigcup \{W : W \text{ is open, } T(\phi) = 0 \text{ for all } \phi \in \mathcal{E}(U, Y) \text{ and } \text{spt } \phi \subseteq W\}.$$

We note that $\text{spt } T$ is compact for all $T \in \mathcal{E}'(U, Y)$. Hence for a compact subset K of U , we may define $\mathcal{E}'_K(U, Y)$ as the set of all $T \in \mathcal{E}'(U, Y)$ such that $\text{spt } T \subseteq K$, so that $\mathcal{E}'_K(U, Y)$ is the union of all such sets. These definitions are similar to the definition of support that we have previously seen in relation to functions on sets, and the notion of compact support is useful here as it is in other results of geometric measure theory.

Definition 1.5. For each compact $K \subseteq U$, we also define

$$\mathcal{D}_K(U, Y) = \{\phi \in \mathcal{E}(U, Y) : \text{spt } \phi \subseteq K\}.$$

Then let $\mathcal{D}(U, Y)$ be the union of the above sets for all such K . We note that every element of $\mathcal{D}(U, Y)$ has compact support.

Definition 1.6. Finally, we also define $\mathcal{D}'(U, Y)$ as the vector space of all continuous linear functions from $\mathcal{D}(U, Y)$ to $\mathbb{R}$. An element of $\mathcal{D}'(U, Y)$ is a **distribution** in U of type Y . $\mathcal{E}'(U, Y)$ is the subset of $\mathcal{D}'(U, Y)$ of compactly supported distributions.

2. Distributions

Now turning our attention of subsets of $\mathbb{R}^n$, we derive some properties of distributions. We will see that these are closely related to integration with respect to the Lebesgue measure $\mathcal{L}^n$. Though not discussed further here, distributions also provide a condition for the existence of Radon measures on a set. We first introduce the notion of smoothing:

Lemma 2.1. *Let $T \in \mathcal{D}'(\mathbb{R}^n, Y)$. Then for all $\epsilon > 0$, there exists $T_\epsilon \in \mathcal{D}'(\mathbb{R}^n, Y)$ such that:*

- (1) $T_\epsilon \rightarrow T$ in $\mathcal{D}'(\mathbb{R}^n, Y)$ as $\epsilon \downarrow 0$.
- (2) $D_j T_\epsilon = (D_j T)_\epsilon$ for all ϵ , where D_j , the partial derivative with respect to the j th variable, is defined on T by $(D_j T)(\phi) = -T(D_j \phi)$.
- (3) For each $\epsilon > 0$, T_ϵ corresponds to a function $g_\epsilon \in \mathcal{E}(\mathbb{R}^n, Y^*)$ mapping $\mathbb{R}^n$ to the dual space of Y .

Proof: Let ϕ be a nonnegative smooth function on $\mathbb{R}^n$, and let $\phi_\epsilon(x) = \epsilon^{-n}\phi(\epsilon^{-1}x)$ for all $x \in \mathbb{R}^n$. Then T_ϵ is given by $T_\epsilon(f) = T(\phi_\epsilon * f)$, where $*$ denotes convolution. The function g_ϵ is given by

$$\langle g_\epsilon(z), y \rangle = T_x[\phi_\epsilon(x - z)y]$$

for all $y \in Y$ and $z \in \mathbb{R}^n$. The subscript x indicates that the argument of T is a function in x . Furthermore, we have

$$T_\epsilon(f) = \int \langle g_\epsilon(z), f(z) \rangle d\mathcal{L}^n z \quad (2.1)$$

for all $f \in \mathcal{D}(\mathbb{R}^n, Y)$. We direct the reader to [Fed96, pp. 346–348] for the detailed proofs of (1) and (3) and [KP08, Lemma 7.3.3] for (2). ■

We then have another result that connects distributions to Lebesgue integration.

Lemma 2.2. *Let U be an open subset of $\mathbb{R}^n$ and $T \in \mathcal{D}'(U, Y)$ a distribution. If A is a connected open subset of U such that*

$$\text{spt } D_j T \subseteq U \setminus A \text{ for all } j = 1, 2, \dots, n.$$

Then there exists a continuous linear function α on Y such that

$$T(f) = \int_U (\alpha \circ f) d\mathcal{L}^n$$

for all $f \in \mathcal{D}(U, Y)$ such that $\text{spt } f \subseteq A$.

Proof: We adapt the proof given in [Fed96, Chapter 4.1.4]. It suffices to show that the statement holds for all A such that $\overline{A}$ is a compact subset of U . We may then take the countable union of such sets to see that it holds for all connected open sets A . Thus we can assume $\text{spt } T$ to be a compact subset of U , since T is contained in the image of $\mathcal{E}'(U, Y)$ in $\mathcal{D}'(U, Y)$ and therefore has compact support. We also assume that $U = \mathbb{R}^n$.

For all $\epsilon > 0$, we define a subset $A_\epsilon = \{x \in A : \text{dist}(x, \mathbb{R}^n \setminus A) > \epsilon\}$. Fixing a point $a \in A$, let C_ϵ be the connected component of A_ϵ containing a . Recalling our definition of g_ϵ from Lemma 2.1 and applying the second statement of the lemma, we have $D_j g_\epsilon(z) = 0$ for all j when $z \in C_\epsilon$. Thus the image of C_ϵ under g_ϵ is a point $a_\epsilon \in Y^*$. Using (2.1), we may then write

$$T(f) = \lim_{\epsilon \rightarrow 0^+} \int \langle a_\epsilon, f(x) \rangle d\mathcal{L}^n x \quad (2.2)$$

for all $f \in \mathcal{D}(\mathbb{R}^n, Y)$ such that $\text{spt } f \subseteq A$. This is because the T_ϵ (which correspond to g_ϵ) converge to T by the first statement of Lemma 2.1.

We now note that for $y \in Y$ and sufficiently small $\delta > 0$, we have

$$\langle a_\delta, y \rangle = \langle g_\delta(a), y \rangle = T_x[\phi_\delta(x - a)y]$$

by our original definition of g_ϵ . Then by (2.2), we can rewrite this as

$$\lim_{\epsilon \rightarrow 0^+} \int \langle a_\epsilon, \phi_\delta(x - a)y \rangle d\mathcal{L}^n x = \lim_{\epsilon \rightarrow 0^+} \langle a_\epsilon, y \rangle.$$

Thus a_δ does not depend on δ for sufficiently small δ , so we may take this function to be our α . ■

3. Currents

With the constructions in the previous sections, we are now ready to define currents.

Definition 3.1. Let U be an open subset of $\mathbb{R}^n$ and m a nonnegative integer. We write the following:

$$\begin{aligned} \mathcal{E}^m(U) &= \mathcal{E}(U, \wedge^m \mathbb{R}^n) & \mathcal{E}_m(U) &= \mathcal{E}'(U, \wedge^m \mathbb{R}^n) \\ \mathcal{D}^m(U) &= \mathcal{D}(U, \wedge^m \mathbb{R}^n) & \mathcal{D}^m(U) &= \mathcal{D}'(U, \wedge^m \mathbb{R}^n) \end{aligned}$$

$\mathcal{E}^m(U)$ is the space of differential forms of degree m and class ∞ on U . $\mathcal{D}^m(U)$ is its subspace of forms with compact support, and an element of $\mathcal{D}_m(U)$ is an m -dimensional **current** in U . $\mathcal{E}_m(U)$ is the subset of $\mathcal{D}_m(U)$ containing the currents with compact support.

We now construct the following operations on currents:

Definition 3.2. Let T be an element of $\mathcal{D}_m(U)$. For some $\phi \in \mathcal{E}^k(U)$, where $k \leq m$, we define an $(m - k)$ dimensional current $T \lrcorner \phi \in \mathcal{D}_{m-k}U$ by

$$(T \lrcorner \phi)(\psi) = T(\phi \wedge \psi)$$

for all $\psi \in \mathcal{D}^{m-k}(U)$. For a smooth p -vectorfield (a map from U to $\wedge_p \mathbb{R}^n$) ξ , define an $m + p$ dimensional current $T \wedge \xi$ by

$$(T \wedge \xi)(\psi) = T(\xi \lrcorner \psi),$$

where $\lrcorner$ denotes the interior product: This is defined by

$$\langle \xi \lrcorner \psi, \alpha \rangle = \langle \psi, \alpha \wedge \xi \rangle.$$

Furthermore, if $m \geq 1$, we define the **boundary** of T :

Definition 3.3. $\partial T \in \mathcal{D}_{m-1}(U)$ is an $m - 1$ dimensional current given by

$$(\partial T)(\psi) = T(d\psi)$$

for all $\psi \in \mathcal{D}^{m-1}(U)$, where d denotes the exterior derivative.

4. The Constancy Theorem

An example of the above constructions is given by the following: Let ξ be an $\mathcal{L}^n$ -measurable p -vector field with $\int_K \|\xi\| d\mathcal{L}^n < \infty$ for all compact $K \subseteq \mathbb{R}^n$. Then we use the Lebesgue measure to define a current $\mathcal{L}^n \wedge \xi \in \mathcal{D}_p(\mathbb{R}^n)$, defined by

$$(\mathcal{L}^n \wedge \xi)(\psi) = \int \langle \xi, \psi \rangle d\mathcal{L}^n$$

for all $\psi \in \mathcal{D}^p(\mathbb{R}^n)$. In particular, we may view $\mathcal{L}^n$ as a 0-dimensional current.

Definition 4.1. We define $\mathbf{E}^n = \mathcal{L}^n \wedge e_1 \wedge \dots \wedge e_n \in \mathcal{D}_n(\mathbb{R}^n)$. This is an n -dimensional current given by

$$\mathbf{E}^n(\phi) = \int \langle \phi(x), e_1 \wedge \dots \wedge e_n \rangle d\mathcal{L}^n x. \quad (4.1)$$

With this definition, we arrive at our main result for this paper.

Theorem 4.2. *Let A be a connected open subset of U , and let $T \in \mathcal{D}_n(U)$ be an n -dimensional current where*

$$\text{spt } \partial T \subseteq U \setminus A.$$

Then there exists $c \in \mathbb{R}$ such that

$$\text{spt}(T - c\mathbf{E}^n \llcorner U) \subseteq U \setminus A.$$

Proof: We proceed according to the outline given in [Fed96, p. 357]. We have the identity $(\partial T) \wedge e_j = (-1)^n D_j T$ (See [Fed96, Chapter 4.1.7]). Thus the support of $D_j T$ is contained in the support of ∂T , so $\text{spt } D_j T \subseteq U \setminus A$. Then if we let $Y = \wedge^n \mathbb{R}^n$, we can apply Lemma 2.2 since $T \in \mathcal{D}'(U, Y)$. Hence there exists a continuous linear function $\alpha \in Y^*$ such that

$$T(f) = \int_U \langle \alpha, f(x) \rangle d\mathcal{L}^n x. \quad (4.2)$$

We now note that for any $\alpha \in Y^*$, since we have a perfect pairing of dual spaces, there exists $x \in \wedge_n \mathbb{R}^n$ such that

$$\langle \alpha, y \rangle = \langle y, x \rangle.$$

But $\wedge_n \mathbb{R}^n$ is spanned by $e_1 \wedge \dots \wedge e_n$, so $x = ce_1 \wedge \dots \wedge e_n$ for some $c \in \mathbb{R}$. Thus rewriting (4.2), we have

$$T(f) = \int_U c \langle f(x), e_1 \wedge \dots \wedge e_n \rangle d\mathcal{L}^n x.$$

Now recalling our definition of $\mathbf{E}^n$ in (4.1), we have $T(f) = \mathbb{E}^n(f)$ for all f such that $\text{spt } f \subseteq A$. Thus we have our desired conclusion. $\blacksquare$

In the case that $A = U$, the above theorem indicates that when the boundary of T vanishes (so that all partial derivatives vanish), T is given by integration.

References

- [Fed96] H. Federer. *Geometric Measure Theory*. Classics in Mathematics. Springer-Verlag Berlin Heidelberg, 1996.
- [KP08] S. G. Krantz and H. R. Parks. *Geometric Integration Theory*. Cornerstones. Birkhäuser Boston, MA, 2008.